GENERAL NOTE ON THE DESIGN

1. BENDS

1.1. R.H. & L. H. STOCK RAILS

FOR THE L.H. TURNOUT,THE L. H. STOCK RAIL IS GIVEN A CURVATURE OF 230587 mm AND THE R. H. STOCK RAIL IS LAID STRAIGHT. THE REVERSE IS THE CASE FOR THE R. H. TURNOUT.

1.2. R.H. & L.H. TONGUE RAILS

THE RAIL IS FIRST BENT THROUGH 45 mm AT POINT 'F' SO THAT AT THE TOE 'E'. THE CENTRE LINE OF HEAD OF TONGUE RAIL IS 9 mm OUTSIDE THE GAUGE LINE OF STOCK RAIL (i.e. TOWARDS CENTRE LINE OF TRACK). AGAIN, AFTER THE PLANING OF THE OUTSIDE HEAD HAS BEEN CARRIED OUT IN STAGES AS EXPLAINED IN 2.2.2. BELOW, THE WHOLE OF THE RAIL LENGTH IS SPRUNG TO A RADIUS OF 232260 mm (CONCAVE OUTSIDE) AND SECURELY CLAMPED IN THIS POSITION TO ENABLE PLANING ON THE INSIDE FOOT AND HEAD TO BE CARRIED OUT AS EXPLAINED IN 2.2.3. & 2.2.4. BELOW.

2. PLANING

2.1. R.H. & L.H. STOCK RAILS

2.1.1. OUTSIDE HEAD & FOOT (i.e. OUTSIDE GAUGE) NO PLANING IS REQUIRED.

2.1.2. INSIDE HEAD

THE HEAD PLANING COMMENCES AT 'A' & ENDS AT 'D', BETWEEN POINTS 'B' & 'C', THE PLANING IS UNIFORM AS SHOWN IN SECTION AT 'C'. THE SIDE OF THE HEAD IS CHAMFERED INWARD AT 1 IN 4 BATTER FROM A POINT 20 mm BELOW TOP OF HEAD. THE PLANING IS RUNOUT STRAIGHT TO ZERO VALUE FROM 'B' TO 'A' AND 'C' TO 'D'.

2.1.3. INSIDE FOOT NO PLANING IS REQUIRED.

2.2. R.H. & L.H. TONGUE RAILS

2.2.1. OUTSIDE FOOT ONLY RELIEF-PLANING IS TO BE DONE AS DESCRIBED IN 2.2.3 BELOW.

2.2.2. OUTSIDE HEAD

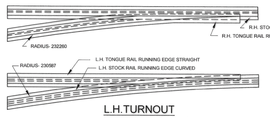
THE PLANING IS IN STAGES. THE PLANING OF THE TOP OF HEAD BEGINS AT 'E' AND ENDS AT 'G'. IN THIS PLANING, THE HEAD IS PLANED DOWN BY 6 mm AT 'E'. THE PLANING OF THE SIDE OF HEAD BEGINS AT 'E' AND ENDS AT 'F'. IN THIS PLANING, THE SIDE HAS AN OUTWARD BATTER OF 1 IN 4 AND THE CORNERS ARE ROUNDED OFF (SEE TEMPLATE 'X'). TEMPLATE 'X' IS DRAWN ON THE END OF THE RAIL AT 'E' AND PLANING GOES STRAIGHT FROM 'E' TO 'G' FOR TOP PLANING AND 'E' TO 'F' FOR SIDE PLANING. THE SECOND PLANING BEGINS AT 'E' AND ENDS AT 'H'. THE TOP OF HEAD IS PLANED DOWN FURTHER BY 16 mm AT 'E' AND THE SIDE OF HEAD IS SET BACK FURTHER (SEE TEMPLATE 'Y'). TEMPLATE 'Y' IS DRAWN ON THE END OF THE RAIL AND PLANING GOES STRAIGHT FROM 'E' TO 'H'.

2.2.3. INSIDE FOOT

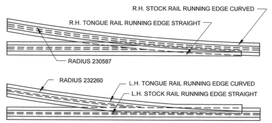
THE PLANING IS IN TWO STAGES . FOR THE FIRST STAGE PLANING, THE RAIL IS SPRUNG TO A RADIUS OF 232260 mm (CONCAVE OUTSIDE) AND SECURELY CLAMPED IN THIS POSTION. THE INNER EDGE OF THE FOOT OF TONGUE RAIL IS CUT BACK TO CLEAR THE WEB OF THE STOCK RAIL. THIS PLANING STARTS AT 31 mm FROM THE CENTRE LINE OF HEAD OF TONGUE RAIL AT 'E' (VIDE TEMPLATE 'Z') AND GOES STRAIGHT FROM 'E' TO 'H'.
FOR THE SECOND STAGE (i.e. RELIEF) PLANING, THE RAIL IS KEPT STRAIGHT AND 48 mm WIDE PORTION OF OUTSIDE FOOT (i.e. ON GAUGE SIDE) AND 17 mm WIDE PORTION OF INSIDE FOOT (i.e. ON NON-GAUGE SIDE) CUT AWAY AS SHOWN IN DETAIL AT 'P' ALONG SIDE.

2.2.4. INSIDE HEAD

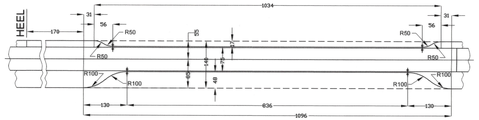
BEFORE PLANING,THE RAIL IS SPRUNG TO RADIUS OF 232260 mm (CONCAVE OUTSIDE) AND SECURELY CLAMPED IN THIS POSITION.
IN THIS PLANING, THE SIDE OF THE HEAD IS GIVEN AN OUTWARD BATTER OF 1 IN 4.34.
THE PLANING BEGINS AT 'E' AND GOES STRAIGHT THROUGH TO 'F' TO FIT THE CUT-HEAD OF STOCK RAIL.



L.H.TURNOUT



R.H. TURNOUT



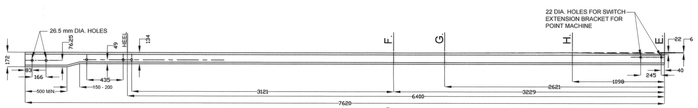
DETAIL AT (P)
(RELIEF PLANNING IN TONGUE RAIL FOOT)



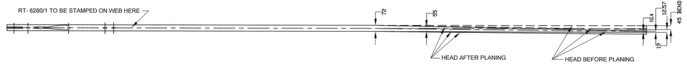
ELEVATION OF R.H. STOCK RAIL (RUNNING EDGE STRAIGHT)



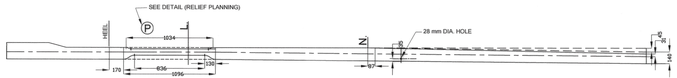
PLAN OF R.H. STOCK RAIL (RUNNING EDGE STRAIGHT) SHOWING PLANING OF HEAD ONLY



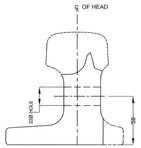
ELEVATION OF R.H. TONGUE RAIL (RUNNING EDGE CURVED) SHOWING TOP PLANING



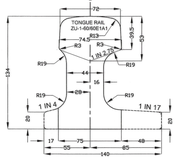
PLAN OF R.H. TONGUE RAIL (RUNNING EDGE CURVED) SHOWING SIDE PLANING OF HEAD ONLY



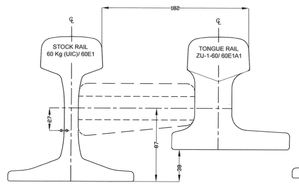
PLAN OF R.H. TONGUE RAIL (RUNNING EDGE CURVED) SHOWING PLANING OF FOOT ONLY



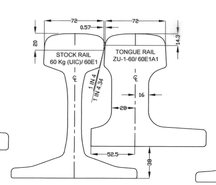
SIDE ELEVATION AT TOE
(SHOWING 22 DIA. HOLES FOR SWITCH EXTENSION BRACKET)



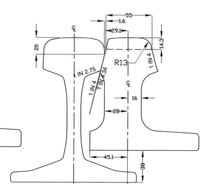
SECTION AT - L



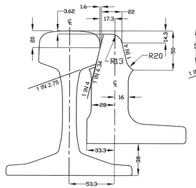
SECTION AT HEEL



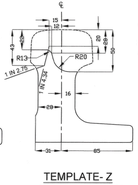
SECTION AT - F
SHOWING DETAILS AT
JUNCTION OF RAIL HEADS (JOH)



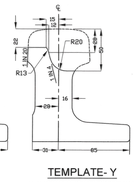
SECTION AT - G
SHOWING 55 mm HEAD



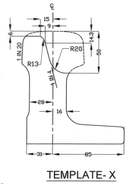
SECTION AT - H
SHOWING 22 mm HEAD



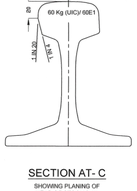
TEMPLATE- Z
SHOWING FINAL SECTION OF
TONGUE RAIL AT ACTUAL TOE 'E'



TEMPLATE- Y
SHOWING SECTION OF TONGUE RAIL
AT 'E' AFTER 2nd PLANING



TEMPLATE- X
SHOWING SECTION OF TONGUE RAIL
AT 'E' AFTER 1st PLANING



SECTION AT- C
SHOWING PLANING OF
HEAD OF STOCK RAIL

4. THE CURVED TONGUE AND STOCK RAILS SHALL BE GIVEN THE CORRECT BEND AS SHOWN IN DRG. No. RDSO/T- 6279 BY THE MANUFACTURER BEFORE DESPATCH FROM HIS WORKS. WHILE INDENTING FOR THE TURNOUTS, THE RAILWAYS SHALL CLEARLY INDICATE THE NUMBER OF L.H. & R.H. TURNOUTS REQUIRED BY THEM. AS THE CURVATURE OF TONGUE & STOCK RAILS GIVEN BY THE MANUFACTURER IS LIKELY TO GET DISTURBED DURING TRANSIT TO THE SITE, THE SAME SHALL BE CHECKED AND RECTIFIED BY THE S.S.E. (P. WAY) WHILE LAYING IN THE TRACK.

3. ALL HOLES ARE 28 mm DIA. EXCEPT WHERE OTHERWISE SHOWN.

2. LATEST ALTERATION NUMBER SHALL BE CHECKED BEFORE PLACING ORDER.

1. ALL DIMENSIONS ARE IN MILLIMETRES.

NOTE

SPECIFICATION	SCALE	ALT:	DESCRIPTION	DATE
IRS. T 10	NOT TO SCALE	02	REVISED & REDRAWN vi) FISH PLATES BOLT HOLES AT BOTH ENDS OF STOCK RAIL REMOVED v) HOLES FOR SWITCH EXTENSION BRACKET ADDED iv) HOLES FOR STRETCHER BAR IN TONGUE RAIL REMOVED iii) HOLE DIA. 22Ø FOR SSD IN TONGUE RAIL IS CORRECTED AS 26Ø ii) DISTANCE OF 1st HOLE FROM JOH GIVEN IN TONGUE RAIL IS CORRECTED AS 3121 mm i) 12R RADIUS REMOVED GIVEN IN TONGUE RAIL AT ATS	03-03-2025

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R. D. S. O.

DETAILS OF TONGUE & STOCK RAILS FOR 6400 mm ZU-1-60 /60E1A1 THICK-WEB CURVED SWITCH B.G. (1673 mm) FOR 60kg (UIC)/ 60E1 ON PSC SLEEPERS

STAGE	ADVANCE	(T)
CAMDLE	24-02-2004	
REVN.		

RDSO/T- 6280/1

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